

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

B.Tech. (IT)

Sem - III (Backlog) University Examination, 2010-11

IT - 202 Data Communication

Date: 18.05.11, Wednesday Time: 10:00 a.m to 01:00 p.m

Maximum Marks: 70

Instructions: (1) All questions are compulsory.

(2) Indicate clearly, the option you attempt along with its respective question number.

(3) Figures to right indicate marks

(4) Rough work is done in the last page of main answer book, don't write anything on the question paper.

SECTION-I

Q1 (a) Attempt all question 05

(1) Explain how the crystal lattice filter acts to provide band-pass filtering. 04

(2) Define transformation ratio in tapped inductor. 01

Q1 (b) A coil has a series resistance of 5Ω , a self capacitance of 7 pF , and an inductance of 1 microH . Determine the effective inductance and effective Q-factor when the coil forms part of a series tuned circuit at 25 MHz . 06

Q2 (a) Write a note on properties of the Fourier Transform. 06

(b) Explain Signal distortion over a communication channel (InterSymbol Interference). 06

OR

Q2 (a) Explain Signal Power and Power spectral Density in detail. 06

(b) Write note on signal energy and Fourier transform. 06

Q3 (a) Write brief notes on the sources of noise, other than thermal, that arises in electronic equipment. Describe how the power Spectral density varies with frequency in each case. 06

P.T.O.

- (b) A Mixer stage has a noise figure of 20dB, and this is preceded by an amplifier that has a noise figure of 9dB and an available power gain of 15dB. Calculate the overall noise figure referred to the input 06

OR

- Q3 (a) Define Noise factor and prove Friis's formula for amplifiers in cascade. 06
- (b) A receiver has a noise figure of 12dB, and it is fed by a low noise amplifier that has a gain of 50dB and a noise temperature of 90K. Calculate the noise temperature of the receiver and the overall noise temperature of the receiving system. 06

SECTION - II

- Q4 (a) Explain hybrid- π equivalent circuit for the BJT with Transconductance, input resistance, collector output capacitances, collector to base capacitance in detail. 04
- (b) A CE amplifier has a tuned circuit in the collector, which resonates at 5MHz with a total tuning capacity of 100pF. The undamped Q factor of the tuned circuit is 150. The amplifier feeds a load resistance of $5K\Omega$, and the output resistance of the transistor is $40K\Omega$. Calculate the voltage gain referred to input terminals and the Miller capacitances at input. The transistor operates at a collector current 500micro A, the collector to base capacitance is 0.6pF. 05
- (c) Write note on application of frequency synthesizer. 02
- Q5 (a) Explain about Adjacent Channel Selectivity. 06
- (b) An AM broadcast receiver has an IF of 465kHz and is tuned to 1000 kHz, and the RF stage has one tuned circuit with a Q of 50.

(a) Find the image frequency (b) Find the image rejection in decibels 06

OR

Q5 (a) What is image rejection? Explain double spotting problem in receiver. 06
(b) Explain method of super heterodyne receiver tracking methods. 06

Q6 (a) Explain how demodulation of amplitude modulation (AM) is done?
Give Suitable circuit and waveform. 05
(b) A Modulating signal consists of a symmetrical triangular wave having zero dc component and peak to peak voltage of 11v. It is used to amplitude modulate a carrier of peak voltage 10v. Calculate the modulation index and the ratio of the side lengths L_1/L_2 of the corresponding trapezoidal pattern. 05
(c) Why crystal oscillator is more effective in communication system over than other oscillator 02

OR

Q6 (a) What is Modulation index; Average power, effective voltage and current in Amplitude modulation? Explain with necessary equation. 05
(b) Explain double sideband suppressed carrier (DSBSC) modulation. 05
(c) What is superheterodyne principle and give its needs? 02
